



$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$$



Prenosna funkcija

$$\sum a_n \frac{d^n}{dt^n} x = \sum b_n \frac{d^n}{dt^n} z$$

$$\sum s^n a_n x(s) = \sum s^n b_n z(s)$$

$$\frac{x(s)}{z(s)} = H(s) = \frac{\sum s^n a_n}{\sum s^n b_n} \quad n \geq m, \quad n \leq 3 \text{ (v praksi)}$$

↳ prenosna funkcija

$f(t)$	$F(s)$
1	$1/s$
e^{at}	$\frac{1}{s-a}$
$f(t)e^{at}$	$F(s-a)$
$\frac{df}{dt}$	$sF(s)$
t	$1/s^2$
t^n	$n! s^{-(n+1)}$
$e^{i\omega t}$	$\frac{1}{1-i\omega}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$\delta(t)$	1

Primer:

1. red

$$\tau \dot{x} + x = z$$

$$\tau s X(s) + X(s) = Z(s)$$

$$H(s) = \frac{1}{1 + \tau s}$$

2. red

$$H(s) = \frac{\omega^2 + 2\zeta\omega s}{s^2 + 2\zeta\omega s + \omega^2}$$

$$z(t) = \delta(t) \rightarrow z(s) = 1$$

$$X(s) = Z(s)H(s) = \frac{1}{1 + \tau s} = \frac{1}{\tau} \frac{1}{s + 1/\tau}$$

$$\rightarrow x(t) = \frac{1}{\tau} e^{-t/\tau}$$

Bodejevi diagrami

$$z(t) = z_0 e^{i\omega t}$$

$$x(t) = x_0 e^{i\omega t} e^{i\phi}$$

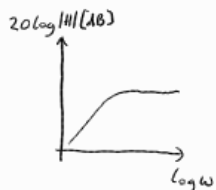
Prenosna funkcija $H = \frac{x}{z}$

velja: $\frac{d}{dt} e^{i\omega t} = i\omega e^{i\omega t} / \mathcal{L}$

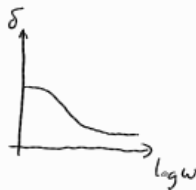
$$s \mathcal{L}\{e^{i\omega t}\} = i\omega \mathcal{L}\{e^{i\omega t}\} \Rightarrow s = i\omega$$

$$H(i\omega) = H(s)$$

$$\frac{x_0}{z_0} = |H(i\omega)|, \quad \tan \phi = \frac{\operatorname{Im}(H(i\omega))}{\operatorname{Re}(H(i\omega))}$$

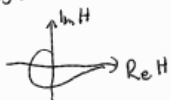


Amplitudni Bod. diagr.



Fazni Bod. diag.

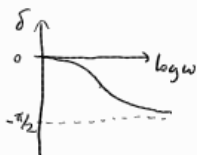
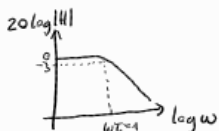
Nyquistov diagram (inženirji)



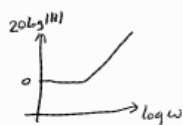
Bodejev diagram - sistem 1. reda

$$H(s) = \frac{1}{1 + \tau s}$$

$$|H(i\omega)| = \frac{1}{\sqrt{1 + \omega^2 \tau^2}}$$

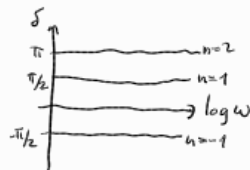
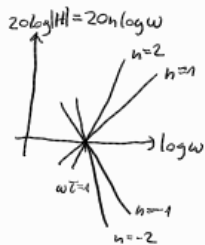


$$H(s) = 1 + \tau s$$

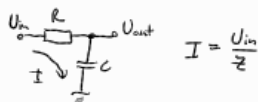


$$H(i\omega) = (i\omega)^n = \omega^n e^{in\pi/2}$$

$$\tan \delta = \frac{\text{Im } H}{\text{Re } H} \rightarrow \delta = n\pi/2$$



L PF

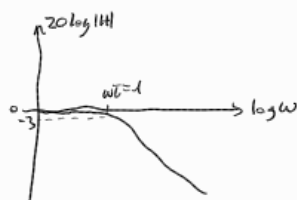


$$I = \frac{U_{in}}{Z}$$

$$Z = R + \frac{1}{i\omega C} = \frac{i\omega RC + 1}{i\omega C} = \frac{1 + i\omega T}{i\omega C}$$

$$U_{out} = Z_C I = \frac{Z_C}{Z} U_{in}$$

$$H = \frac{U_{out}}{U_{in}} = \frac{Z_C}{Z} = \frac{1}{1 + i\omega T}$$



HPF

$$I = \frac{U_{in}}{Z} = U_{in} \frac{i\omega C}{1 + i\omega RC}$$

$$Z = \frac{1}{i\omega C} + R$$

$$U_{out} = I R = U_{in} \frac{i\omega T}{1 + i\omega T}$$

$$H = \frac{U_{out}}{U_{in}} = \frac{i\omega T}{1 + i\omega T}$$

$$|H| = \frac{\omega T}{\sqrt{1 + \omega^2 T^2}}$$

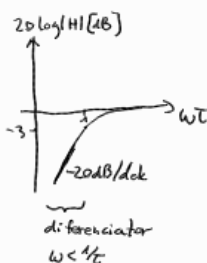
$$\omega T \ll 1 \rightarrow 20 \log |H| = 20 \log \omega T \rightarrow -\infty$$

$$\omega T \gg 1 \rightarrow 20 \log |H| \rightarrow 0$$

$$\omega T = 1 \rightarrow 20 \log \frac{1}{\sqrt{2}} = -3$$

$$H_{diff} = i\omega$$

$$\rightarrow H = H_{diff} T$$



2. red

$$H(s) = \frac{1}{s^2 \tau^2 + \frac{2\xi s}{\omega_0} + 1} = \frac{1}{(1 - \frac{\omega^2}{\omega_0^2}) + 2\xi \frac{i\omega}{\omega_0}}$$

$$s = i\omega$$

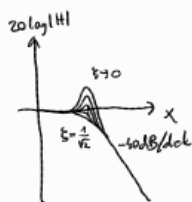
$$\frac{\omega}{\omega_0} = x$$

$$|H(i\omega)| = \left((1-x^2)^2 + 4\xi^2 x^2 \right)^{-1/2}$$

$$x \ll 1: |H| \rightarrow 1, 20 \log |H| \rightarrow 0$$

$$x \gg 1: |H| \rightarrow x^{-2}, 20 \log |H| \rightarrow -40 \log x$$

$$x = 1: |H| \rightarrow (4\xi^2)^{-1/2}, 20 \log |H| \rightarrow \frac{20}{\xi} \log \frac{1}{4\xi^2} \rightarrow -10 \log (4\xi^2)$$



Fazni del

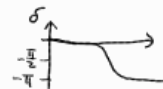
$$H(i\omega) = \frac{(1 - \frac{\omega^2}{\omega_0^2}) - 2\xi i \frac{\omega}{\omega_0}}{(1 - \frac{\omega^2}{\omega_0^2})^2 + 4\xi^2 (\frac{\omega}{\omega_0})^2}$$

$$\tan \delta = \frac{\text{Im } H}{\text{Re } H} = \frac{-2\xi \frac{\omega}{\omega_0}}{1 - \frac{\omega^2}{\omega_0^2}}$$

$$\omega \rightarrow 0: \tan \delta \rightarrow 0, \delta \rightarrow 0$$

$$\omega \rightarrow \infty: \tan \delta = \frac{2\xi}{x} \rightarrow \delta \rightarrow -\pi$$

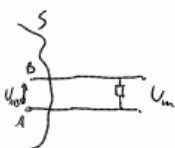
$$x = 1: \tan \delta = -\frac{2\xi}{1-x} \rightarrow -\infty \rightarrow \delta = -\frac{\pi}{2}$$



Vpliv senzorja na opazovani sistem

- Moze P, ki jo senzor črpa iz S za delovanje

npr. voltmeter

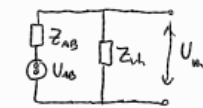


$$P = \frac{U_m^2}{Z_k} \rightarrow \text{vhodna upornost voltmetra}$$

$$U_k < U_{AB}$$

Théveninov izrek

Vsako el. vezje, ki ga sestavljajo le linearni elementi + napetostni vir ter ga opazujemo v točkah A in B, lahko nadomestimo z virom z napetostjo U_{AB} in notranjo upornostjo Z_{AB} .



$$Z = Z_{AB} + Z_{LH}$$

$$I = \frac{U_{AB}}{Z_{AB} + Z_{LH}}$$

$$U_{LH} = I Z_{LH} = \frac{Z_{LH} U_{AB}}{Z_{AB} + Z_{LH}}$$

$$P = \frac{U_{LH}^2}{Z_{LH}} = \frac{U_{AB}^2 Z_{LH}}{(Z_{AB} + Z_{LH})^2}$$

$$P = \max?$$

$$\frac{dP}{dZ_{LH}} = \dots$$

$$\rightarrow Z_{AB} = Z_{LH}$$

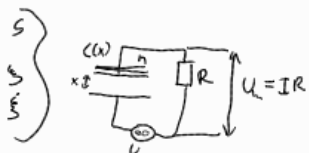
$$P_{\max} = \frac{U_{AB}^2}{4 Z_{AB}}$$

$$P = 4 P_{\max} \frac{Z_{LH}}{Z_{AB}} \left(1 + \frac{Z_{LH}}{Z_{AB}}\right)^{-2} \sim 4 P_{\max} \frac{Z_{AB}}{Z_{LH}} \quad \text{if } Z_{LH} \gg Z_{AB}$$

$$\frac{P}{P_{\max}} = 4 \frac{Z_{AB}}{Z_{LH}}$$

$\rightarrow Z_{LH}$ mora biti zin večji, da bo upliv zin manjši

Kondenzatorski senzor premika



$$C = \frac{\epsilon \epsilon_0 S}{x} \quad dC = -\frac{\epsilon \epsilon_0 S}{x^2} dx = -C \frac{dx}{x}$$

$$\frac{dC}{dt} = -\frac{C}{x_0} \frac{dx}{dt}$$

$$F_c = \frac{1}{2} \epsilon E^2 \quad (\text{silna ene plošče na drugo})$$

$$W_c = \frac{1}{2} C U_c^2$$

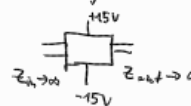
$$dW_c = \frac{1}{2} U_c^2 dC = -\frac{1}{2} U_c^2 \frac{C}{x_0} dx$$

$$-\frac{dW_c}{dx} = F_c = \frac{1}{2} U_c^2 \frac{C}{x_0} = \frac{1}{2} e \underbrace{\frac{1}{C} \frac{C}{x_0}}_E = \frac{1}{2} e E$$

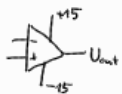
$$P = \frac{dW}{dt} = \frac{1}{2} e E \dot{x} = \frac{1}{2} C U_c^2 \frac{\dot{x}}{x_0}$$

Želimo majhen C, U_c in velik $\dot{x}$
 $U_m = IR = R \frac{dC}{dt} = R(C U_c)' = R U_c \frac{C}{x_0} \dot{x}$
 Želimo velik $U_m \Rightarrow$ velike C, U_c , majhen $\dot{x}$

$\Rightarrow$ predajačevalnik



Operacijski ojačevalnik



$$A_{DC} \sim 10^6 = f(f_T, \dots)$$

$$A(s) = A_{DC} \cdot H(s)$$

$$U_{out} = A(U_+ - U_-)$$

Negativna povratna zanka



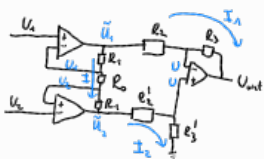
Fal fajr, ker U_{out} ne črpa moči od U_{in}

$$U_{out} = A(U_{in} - F U_{out})$$

$$\frac{U_{out}}{U_{in}} = H = \frac{A}{1 + AF} \approx \frac{1}{\frac{1}{A} + F} \approx \frac{1}{F} \quad A \gg 1$$

Instrumentacijski ojačevalnik

• diferencialni predajačevalnik, ki ga sestavljajo 3 OPAMP (operacijski ojačevalnik) $\Rightarrow$



$$I = \frac{U_1 - U_2}{R_0} = \frac{\tilde{U}_1 - U_2}{R_1} = \frac{U_1 - \tilde{U}_2}{R_1}$$

$$\Rightarrow U_1 + U_2 = \tilde{U}_1 + \tilde{U}_2$$

$$2 \frac{U_1 - U_2}{R_0} = \frac{\tilde{U}_1 - U_1}{R_1} + \frac{U_1 - \tilde{U}_2}{R_1} = \frac{\Delta \tilde{U} - \Delta U}{R_1}$$

$$\Delta \tilde{U} = \left(\frac{2 R_1}{R_0} + 1 \right) \Delta U$$

$$I_1 = \frac{\tilde{U}_1 - U}{R_2} = \frac{U - U_{out}}{R_3} \rightarrow \tilde{U}_1 \frac{R_3}{R_2} - U \frac{R_3}{R_2} - U_{out} = \frac{R_3}{R_2} \tilde{U}_1 - \left(1 + \frac{R_3}{R_2}\right) U$$

$$I_2 = \frac{\tilde{U}_2 - U}{R_2} = \frac{U - 0}{R_3} \rightarrow U = \frac{R_3}{R_2 + R_3} \tilde{U}_2 = k' \tilde{U}_2$$

$$U_{out} = \frac{R_3}{R_2} \tilde{U}_1 + \left(\frac{R_3}{R_2} + 1 \right) k' \tilde{U}_2 = -\frac{R_3}{R_2} \left(\tilde{U}_1 + \left(1 + \frac{R_2}{R_3}\right) k' \tilde{U}_2 \right)$$

želimo, da je 1

$$1 = \left(1 + \frac{R_2}{R_3}\right) \left(1 + \frac{R_2}{R_3}\right)^{-1}$$

$$\frac{R_2}{R_3} = \frac{R_2'}{R_3'}$$

$$\rightarrow U_{out} = -\frac{R_3}{R_2} \left(\frac{2 R_1}{R_0} + 1 \right) \Delta \tilde{U}$$

Rezultat:

common mode rejection ratio

$$CMRR = \frac{\Delta U}{\Delta U_{CM}} \sim 10^6 \Rightarrow \text{offset se izniči, ga s korajdn ne vidimo}$$

• razlika se ojačuje precej več kot vsota

$$\left(1 + \frac{R_2}{R_3}\right) k' = \frac{1 + \frac{R_2}{R_3}}{1 - \frac{R_2}{R_3}}, \quad \epsilon \rightarrow 0$$

$$U_{out} = -\frac{R_3}{R_2} \left(\tilde{U}_1 + \frac{1 + \frac{R_2}{R_3}}{1 - \frac{R_2}{R_3}} \tilde{U}_2 \right) \sim -\frac{R_3}{R_2} \Delta \tilde{U}$$